

Week 4 Lecture Note

Topic: Testing the Saved Model and Making Predictions

1. Objectives

- (1) Understand what **prediction** means in AI and load the model saved in Week 3
- (2) Test the model on unseen images and see whether the model predicts **Normal** or **Pneumonia**
- (3) Understand how AI can help in medical image analysis.

2. Key Concepts

(1) What is Inference (or Prediction)?

Inference means using a trained model to make a prediction.

For example:

- input: a new chest X-ray image
- output: **Normal** or **Pneumonia**

(2) Why Use a Saved Model?

Because the model has already learned in Week 3. So instead of training again, we can:

- load the saved model
- use it immediately
- test it on images

(3) Confidence / Probability

The model output is often a number between 0 and 1. Example:

- 0.10 → likely Normal
- 0.85 → likely Pneumonia

A simple rule:

- below 0.5 → Normal
- 0.5 or above → Pneumonia

(4) Real-World Meaning

You can connect this to medical applications. AI does not replace doctors, but it can help detect patterns quickly and support medical decision-making.

We built a complete AI medical imaging project. First, we explored the data. Then we built a CNN, trained it, saved it, and finally used it to predict new chest X-ray images.